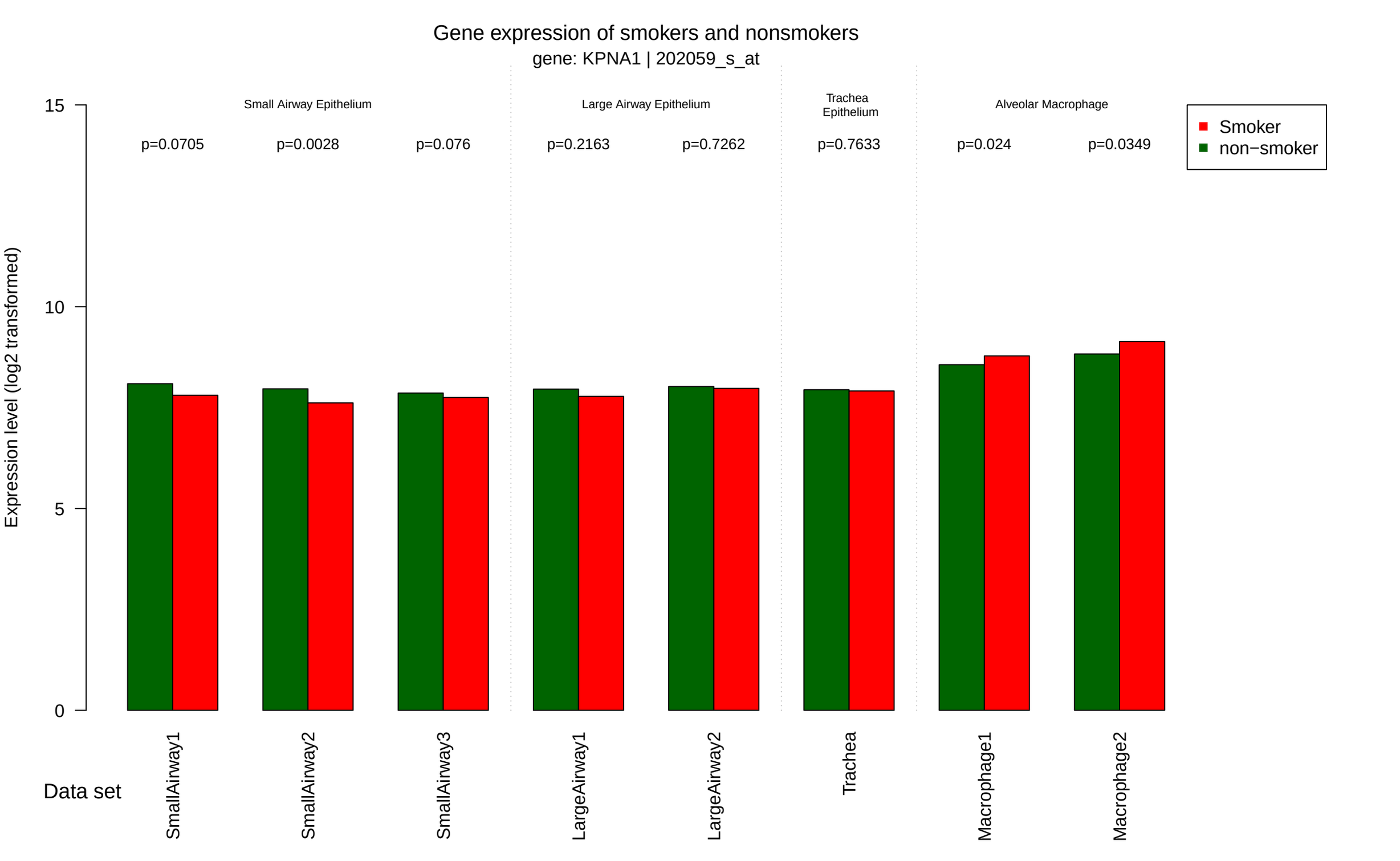
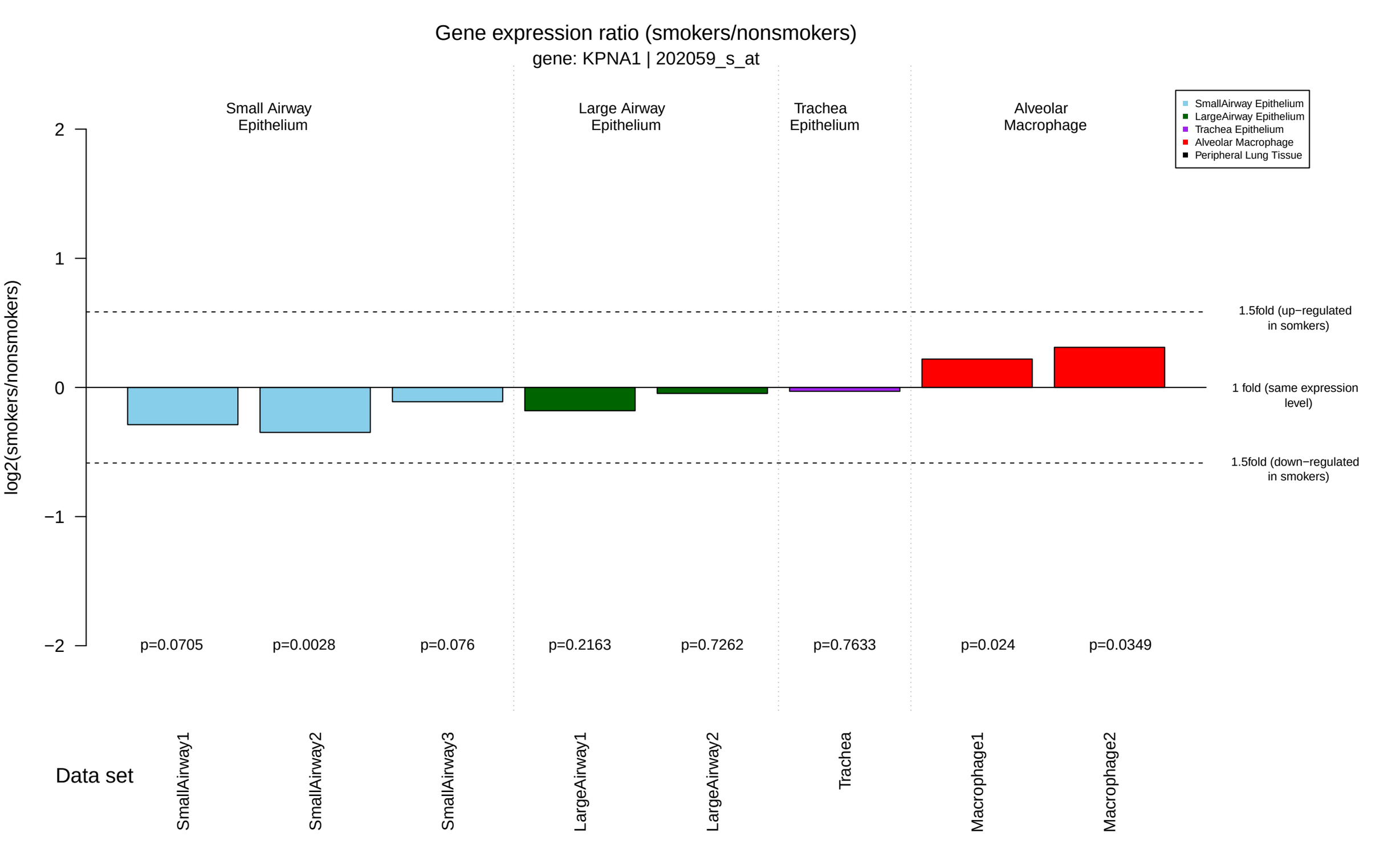
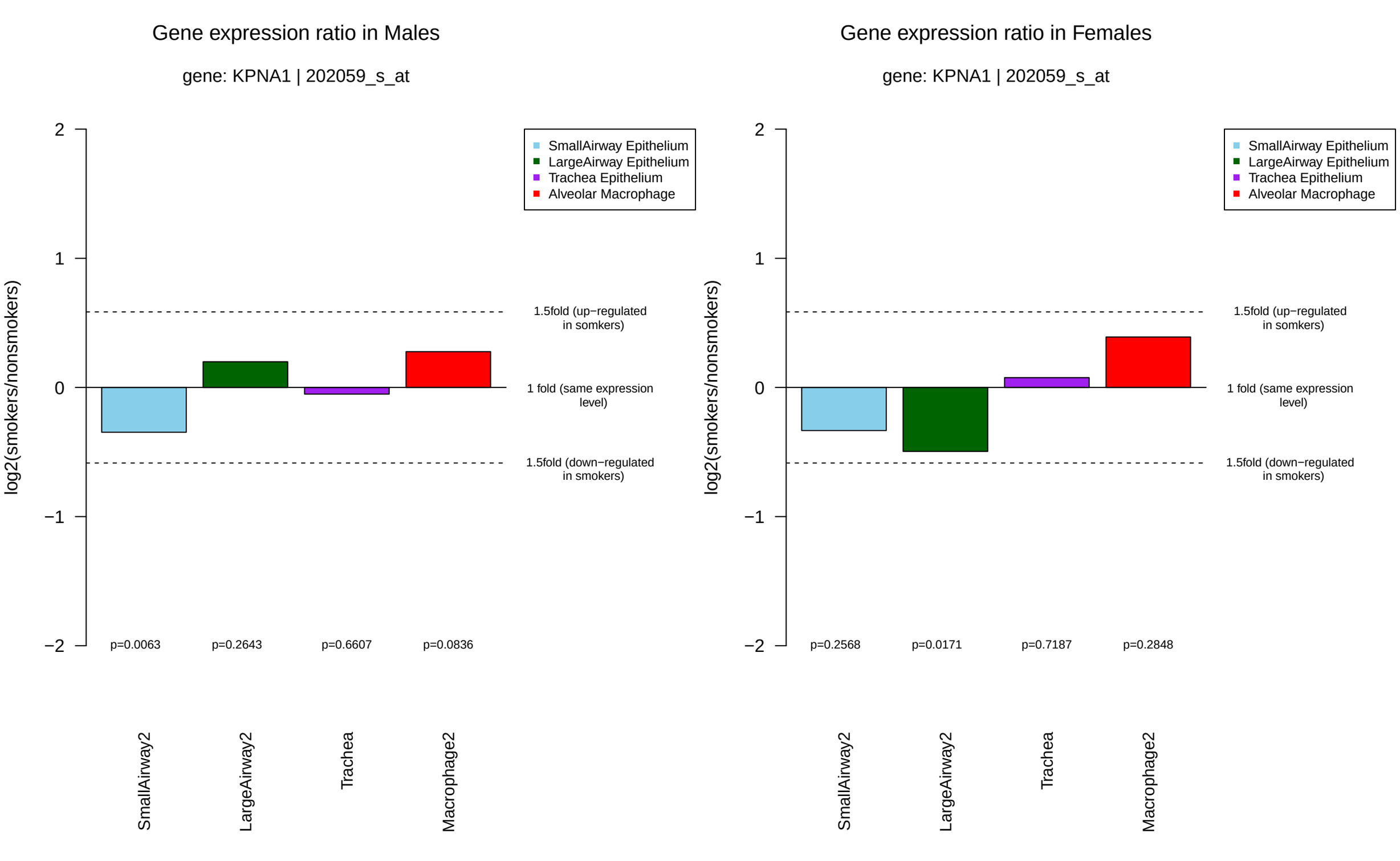
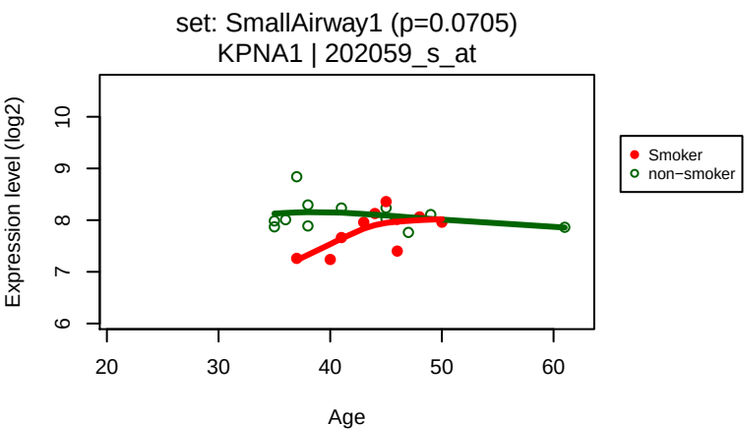
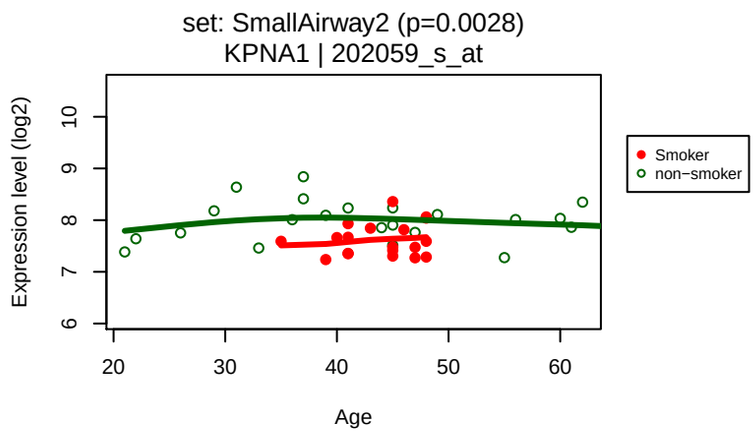
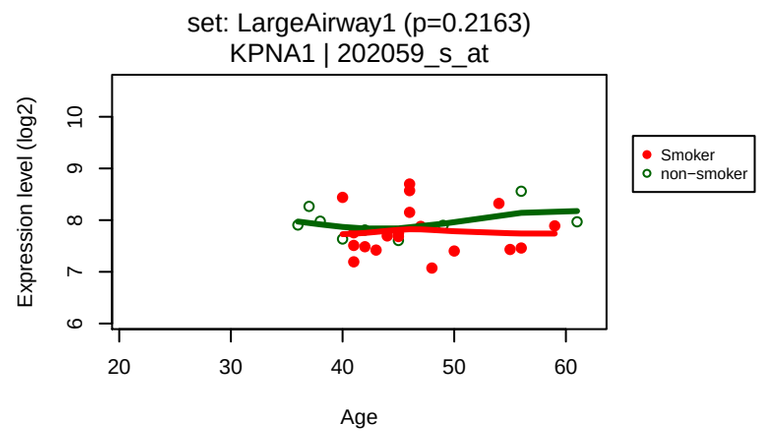

Gene expression of smokers and nonsmokers
gene: KPNA1 | 202059_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0705
p=0.0028
p=0.076
p=0.2163
p=0.7262
p=0.7633
p=0.024
p=0.0349
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

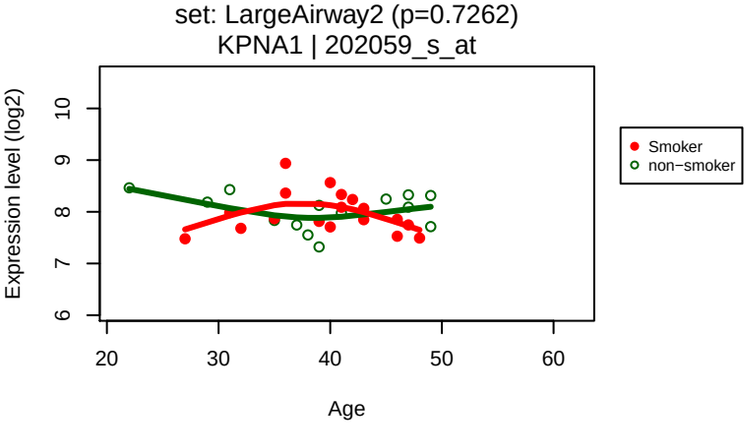

Gene expression ratio (smokers/nonsmokers)
gene: KPNA1 | 202059_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0705
p=0.0028
p=0.076
p=0.2163
p=0.7262
p=0.7633
p=0.024
p=0.0349
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

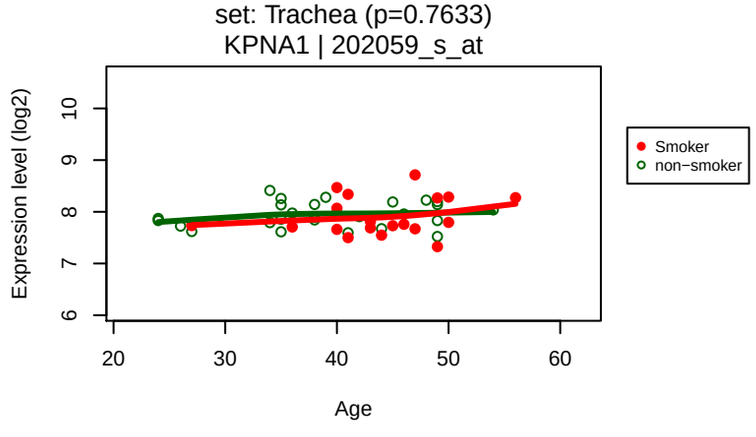

Gene expression ratio in Males
gene: KPNA1 | 202059_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0063
p=0.2643
p=0.6607
p=0.0836
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KPNA1 | 202059_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2568
p=0.0171
p=0.7187
p=0.2848
SmallAirway2
LargeAirway2
Trachea
Macrophage2

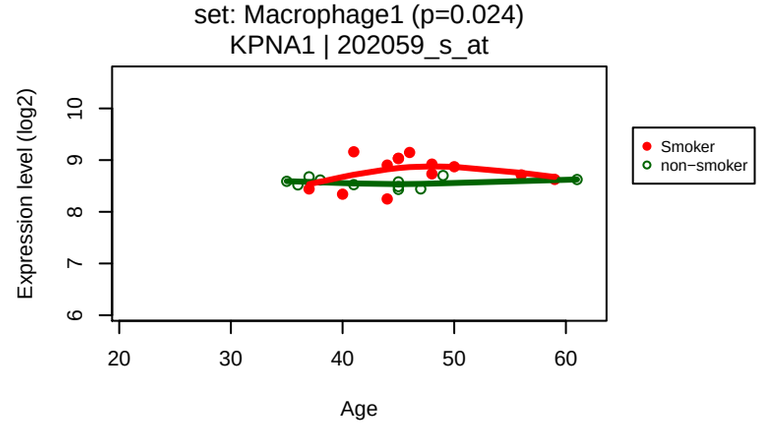

set: SmallAirway1 (p=0.0705)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker

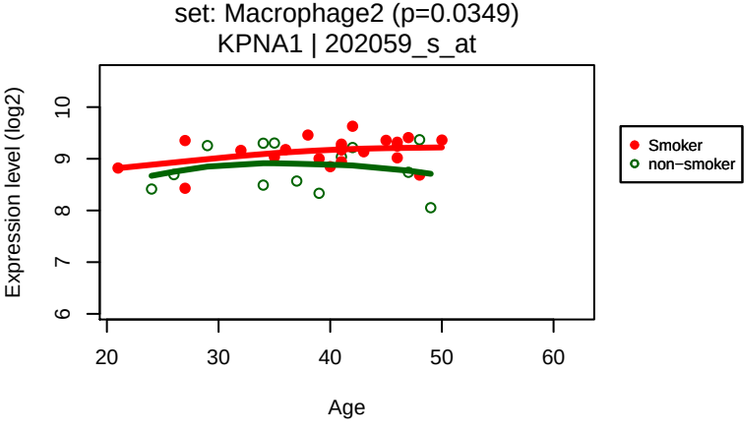

set: SmallAirway2 (p=0.0028)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker

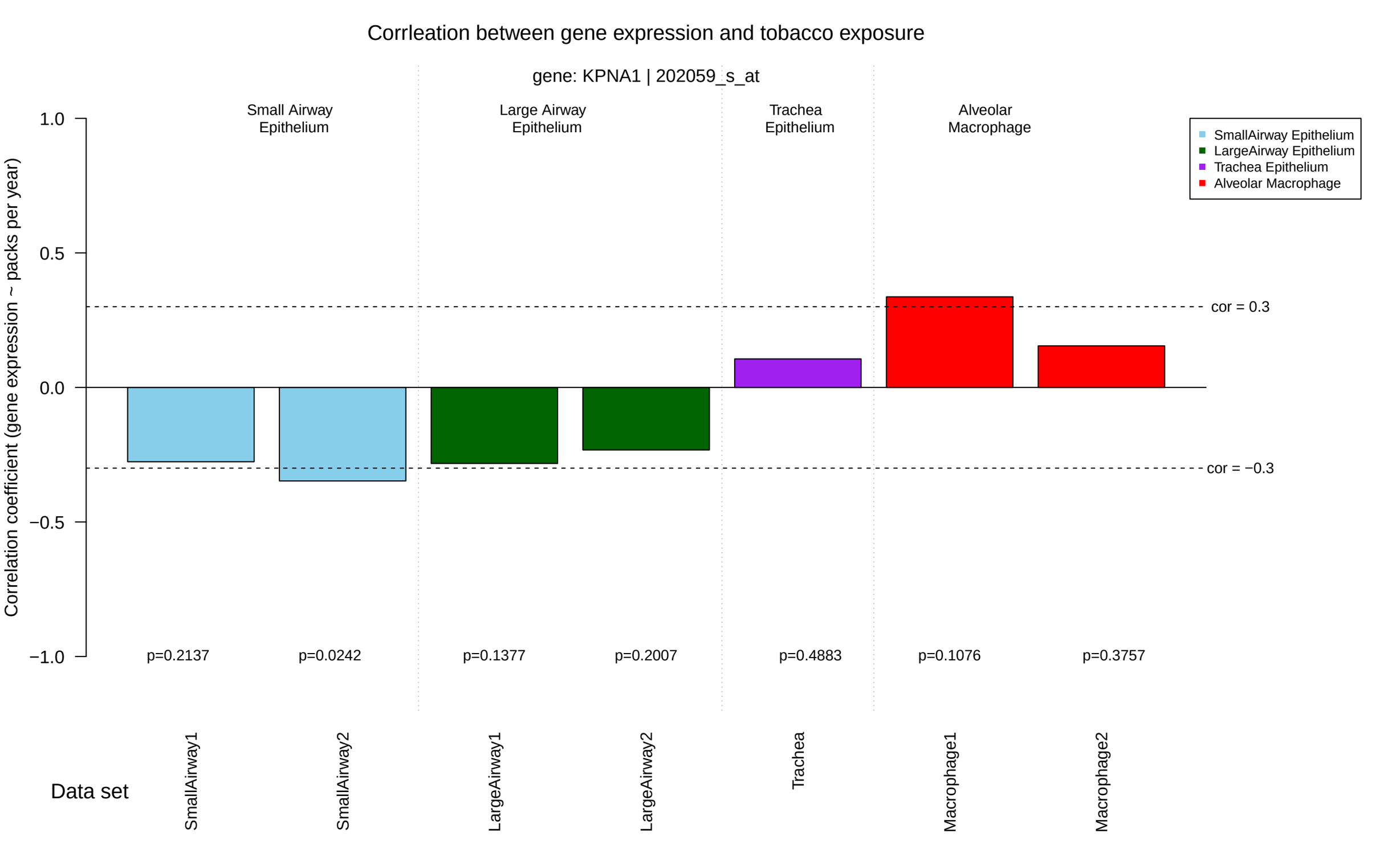

set: LargeAirway1 (p=0.2163)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7262)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7633)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.024)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0349)
KPNA1 | 202059_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KPNA1 | 202059_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2137
p=0.0242
p=0.1377
p=0.2007
p=0.4883
p=0.1076
p=0.3757
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2